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Knowledge sharing platform

Group Details

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- Asst.Prof. Khodidas Vaghela
- Group no: 7

1. Project Profile

1.1 . Existing System

- In most colleges, students form study groups informally using WhatsApp or other social media.

- There is no centralized platform to:
    - Join topic-based groups
    - Share study materials
    - Collaborate with peers
- Notes are shared in unorganized ways (messages, random folders).

1.2. Proposed System

- A website where students can:
  - Create or join study groups
  - Chat or collaborate
  - Upload and download study materials (notes, PDFs)
  - Access premium features
- Admins can:
  - Moderate content, add premium content

1.3. Development Tools and Technology

Front-end

- HTML
- CSS
- JavaScript

Back-end

- Python
- Django

Users Wise Role

Student

- Log-in
- Registration
- Join or create study groups
- Upload & download notes

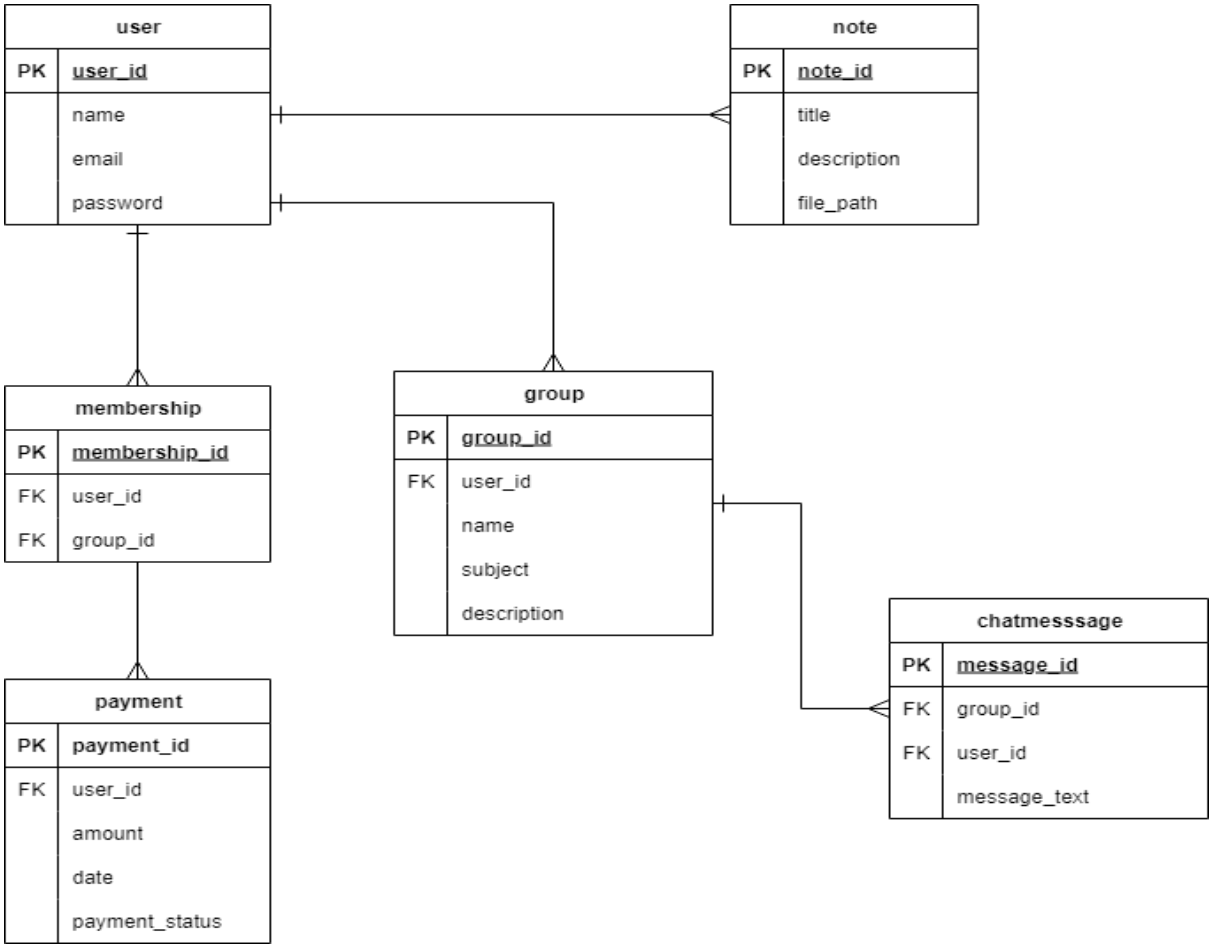
Users Wise Role

Admin:

- Approve user-generated content
- Manage users
- Manage groups
- Mange Reports
- Manage sharing and downloading materials

ER Diagram

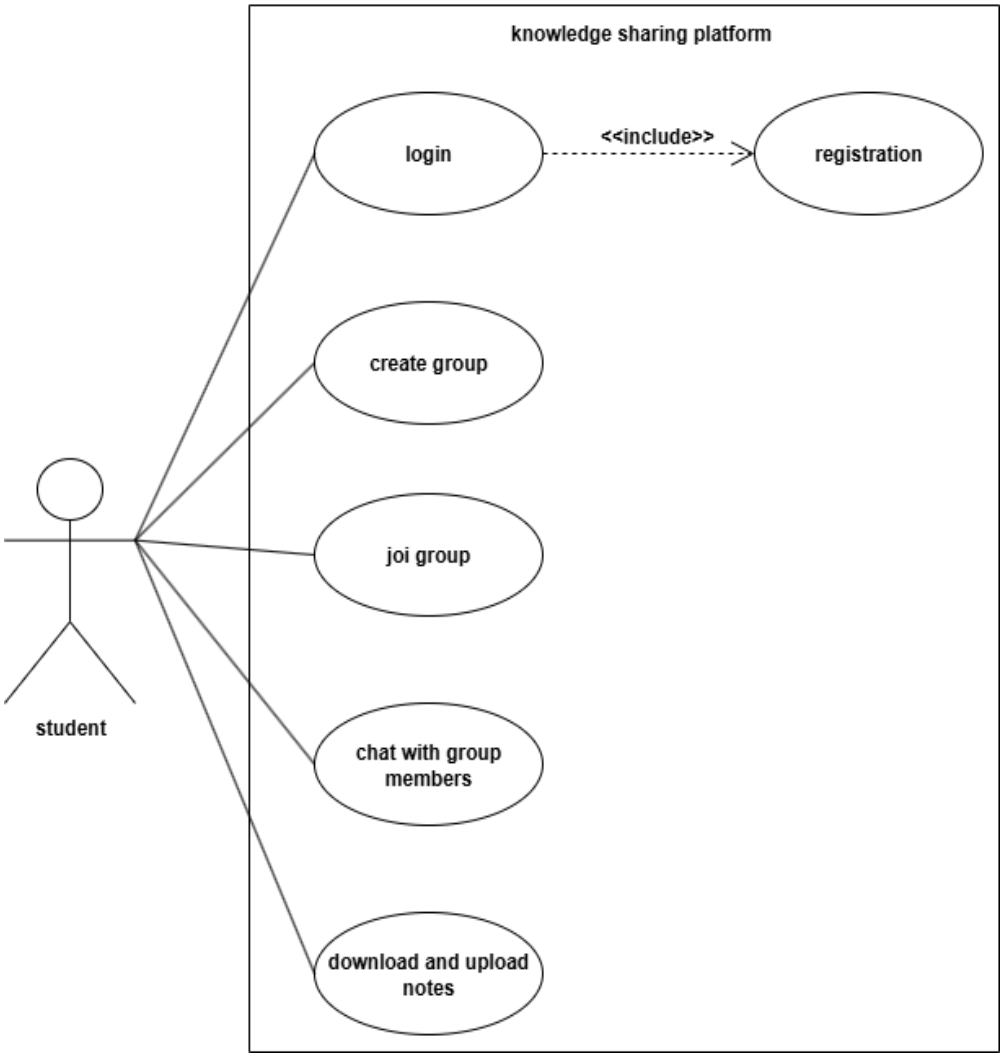
An ER Diagram (Entity-Relationship Diagram) is a visual representation of the data and relationships between entities in a database. It is used in database design to show how data is structured and how different entities (such as users, products, or transactions) are related to each other.



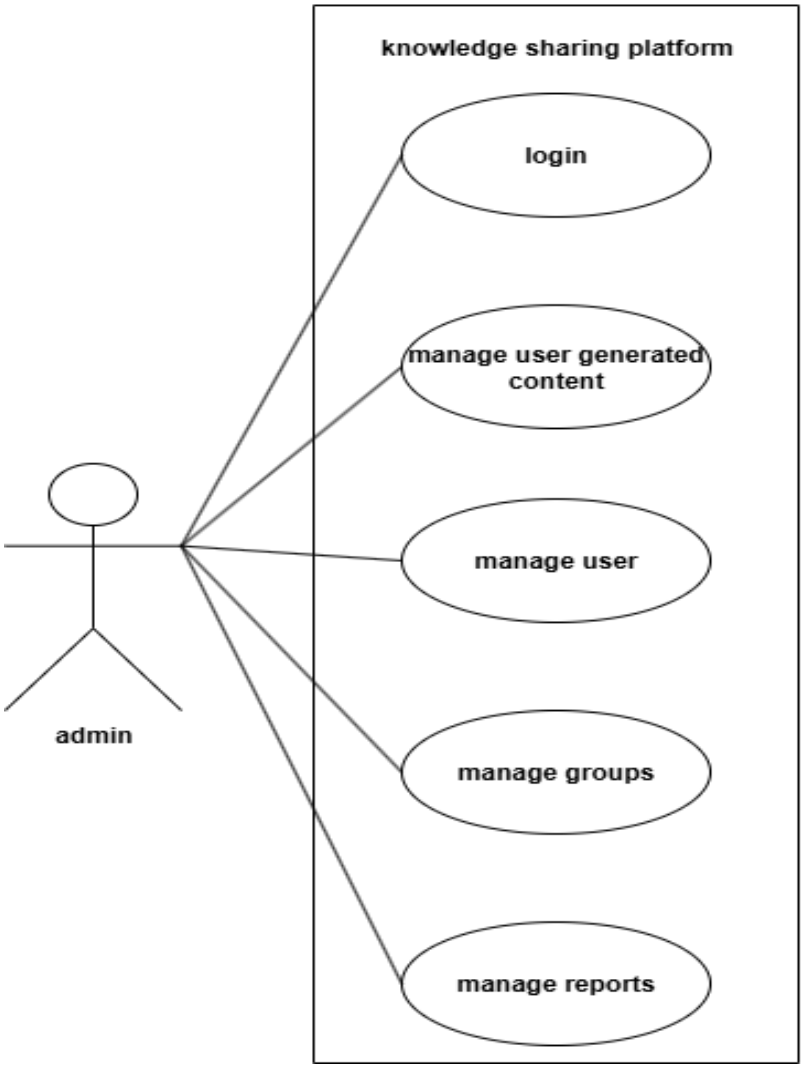
UML Diagram

A Use Case Diagram is a type of UML (Unified Modelling Language) diagram that visually represents the interaction between users (actors) and the system to achieve specific goals. It shows what the system does from the user's point of view, rather than how it is done.

user:



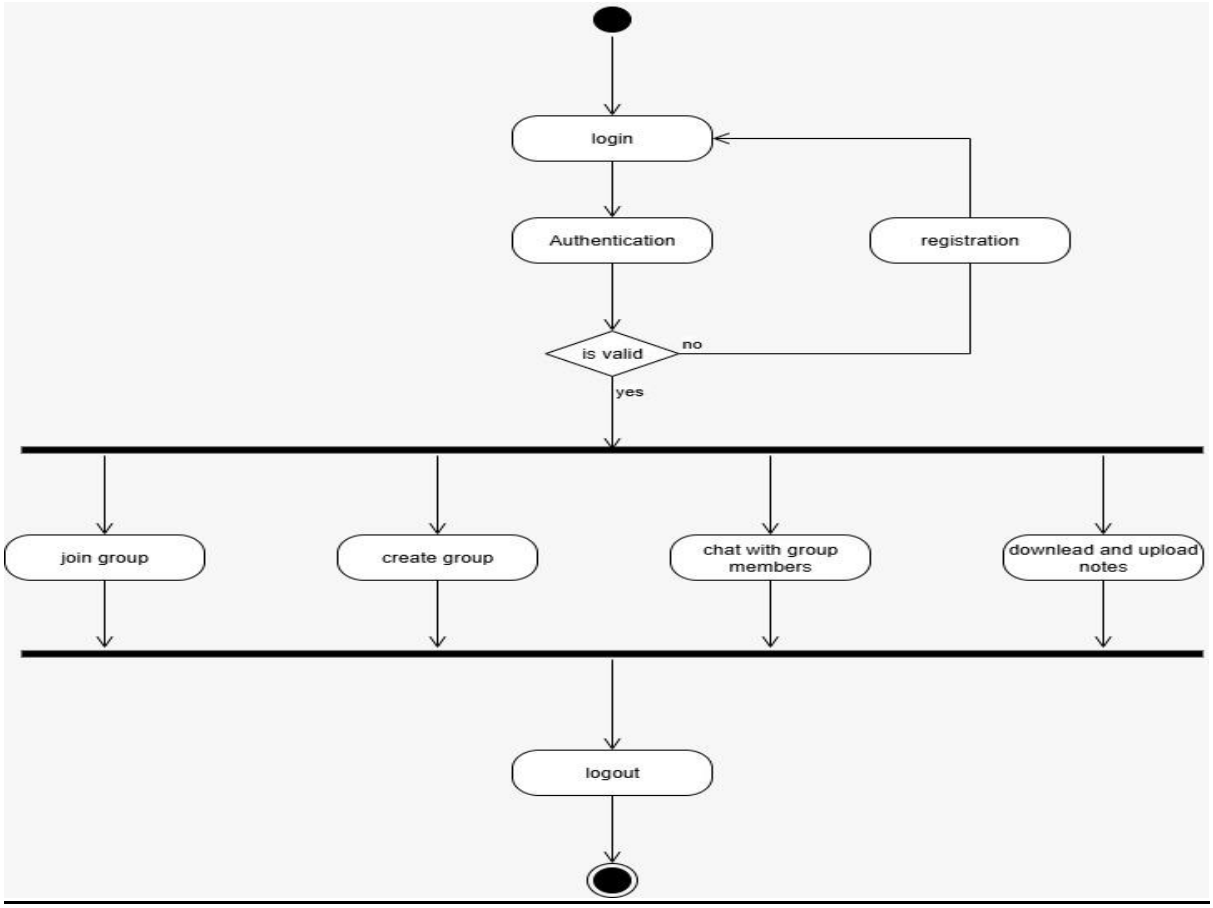
Admin:


knowledge sharing platform
login
manage user generated content
manage user
manage groups
manage reports
admin

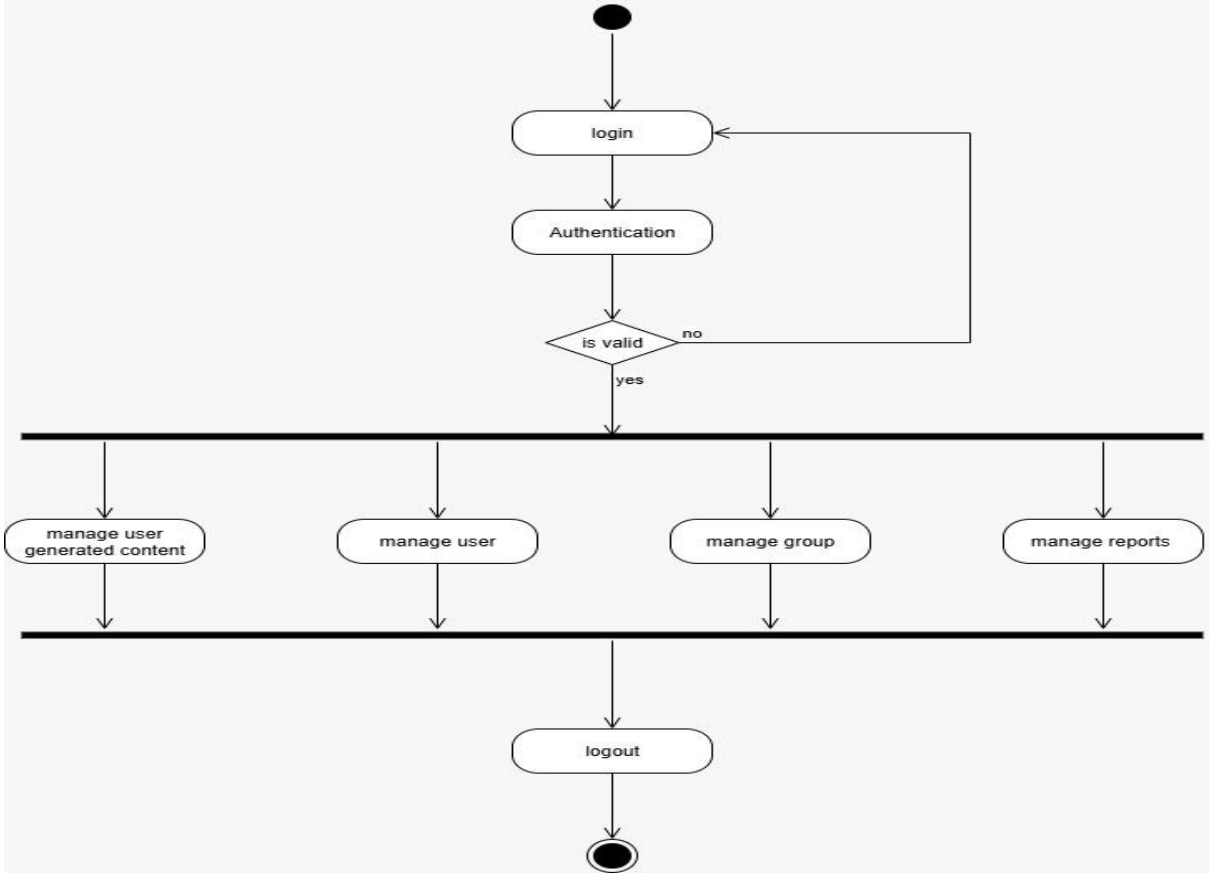
Activity diagram

An Activity Diagram is a type of UML (Unified Modelling Language) diagram that represents the flow of activities or actions in a system. It shows how a process starts, the sequence of steps involved, decisions made, and how it ends.

user:



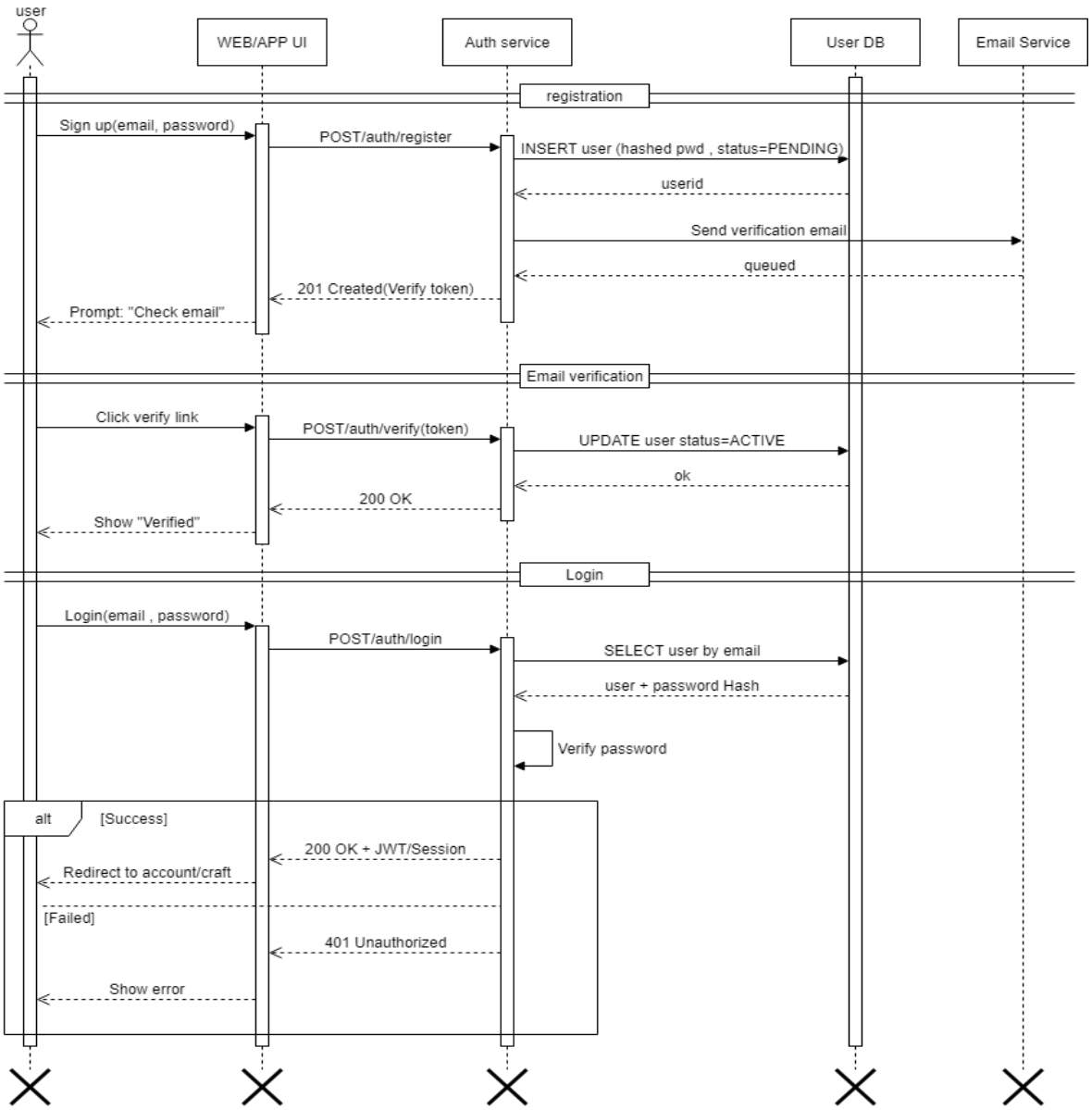
Admin:

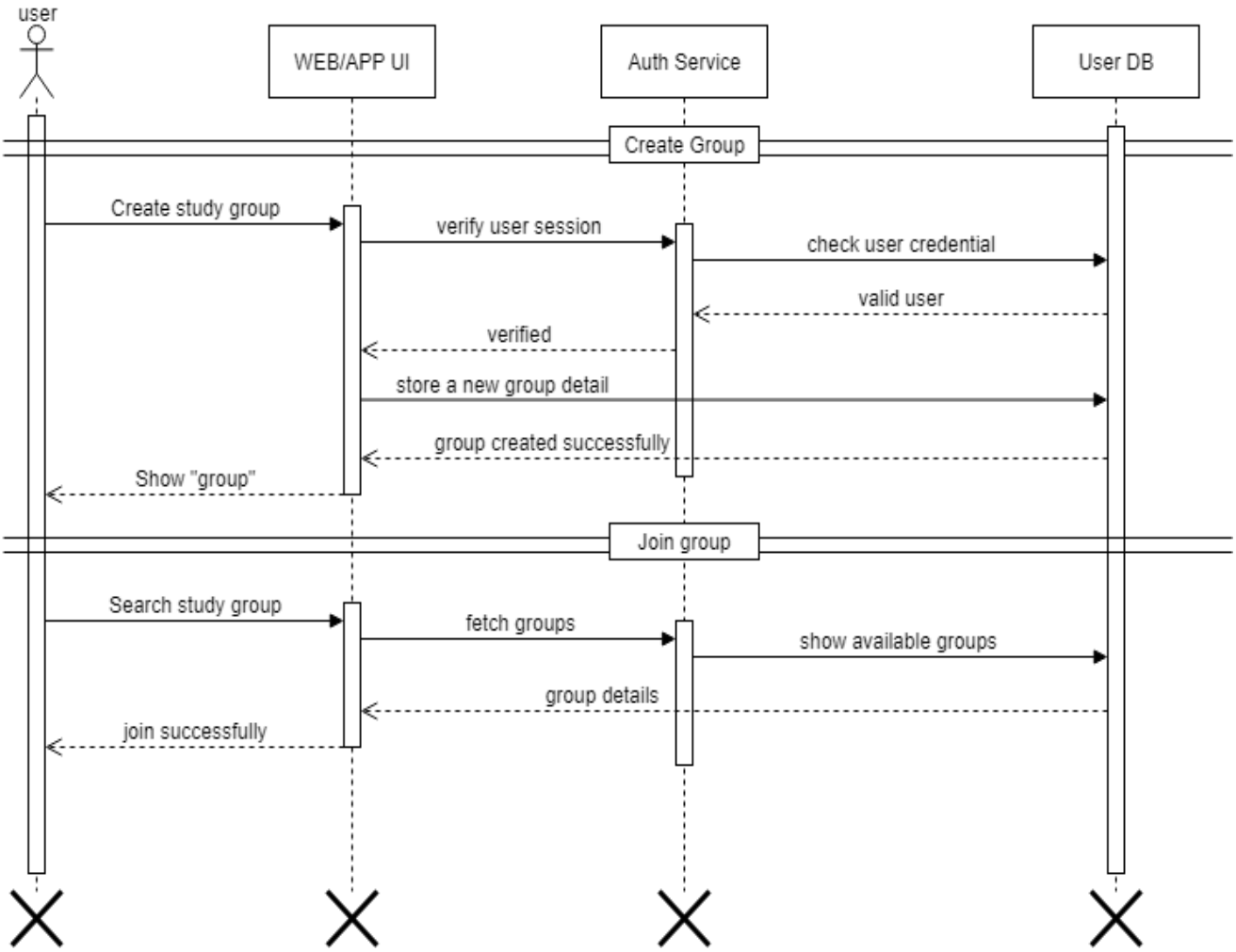

login
Authentication
is valid
no
yes
manage user generated content
manage user
manage group
manage reports
logout

Sequence Diagram

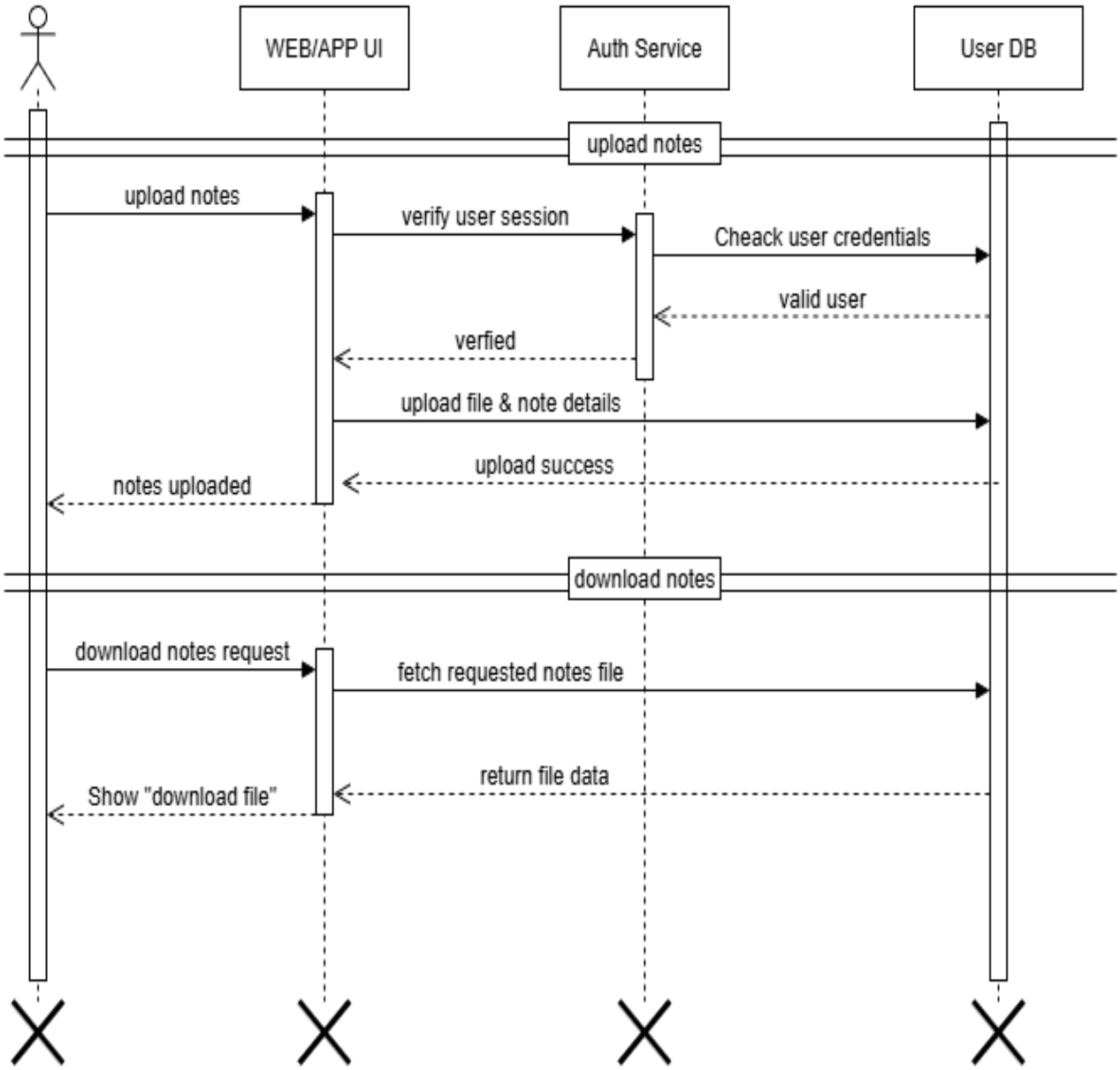
A Sequence Diagram is a type of UML (Unified Modelling Language) diagram that shows the interaction between objects or components in a system in a time sequence. It illustrates how messages are passed between different parts of the system to complete a specific process or function.

user:

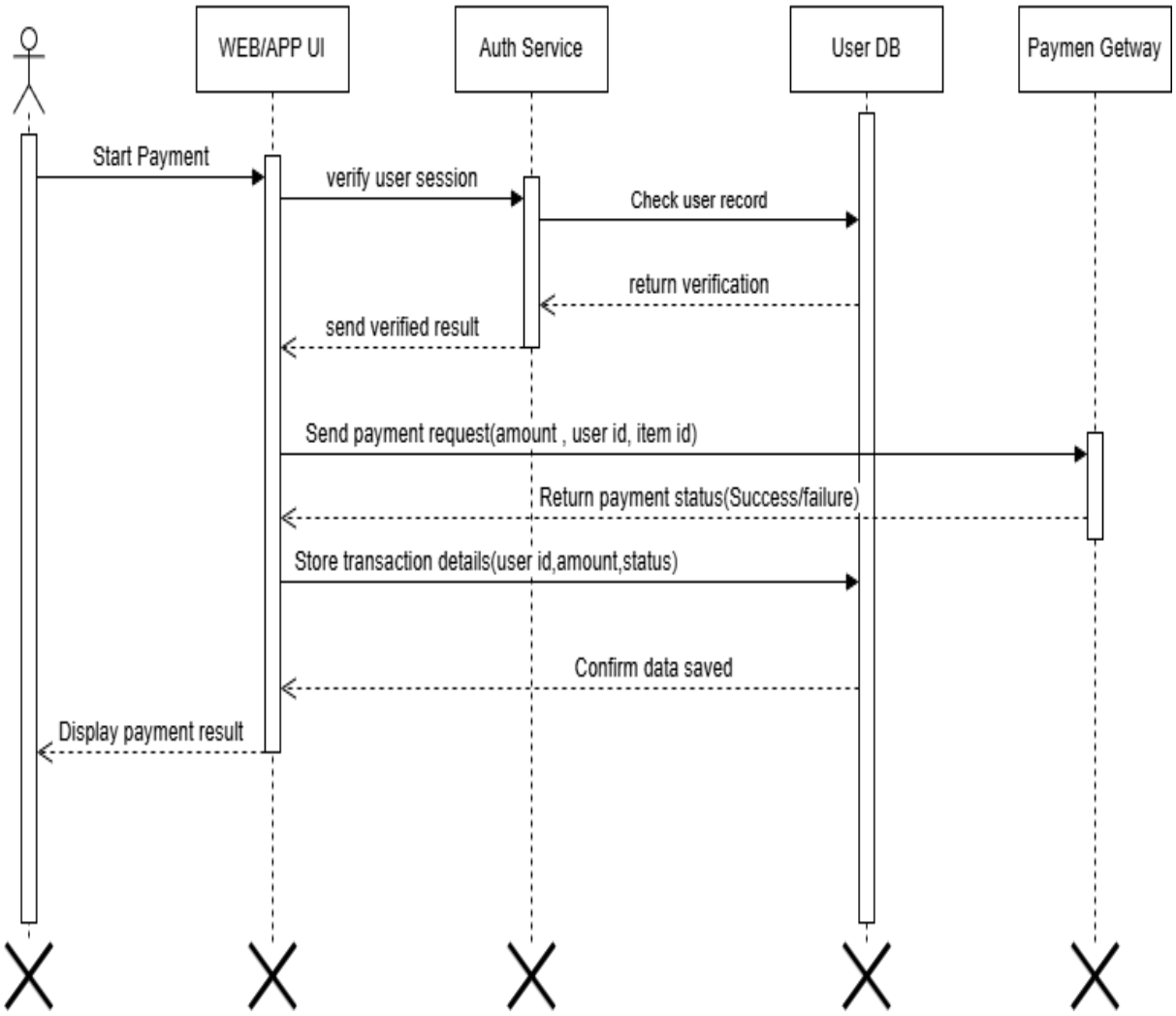



user
WEB/APP UI
Auth Service
User DB
Create Group
Create study group
verify user session
check user credential
valid user
verified
store a new group detail
group created successfully
Show "group"
Join group
Search study group
fetch groups
show available groups
group details
join successfully

upload and download notes

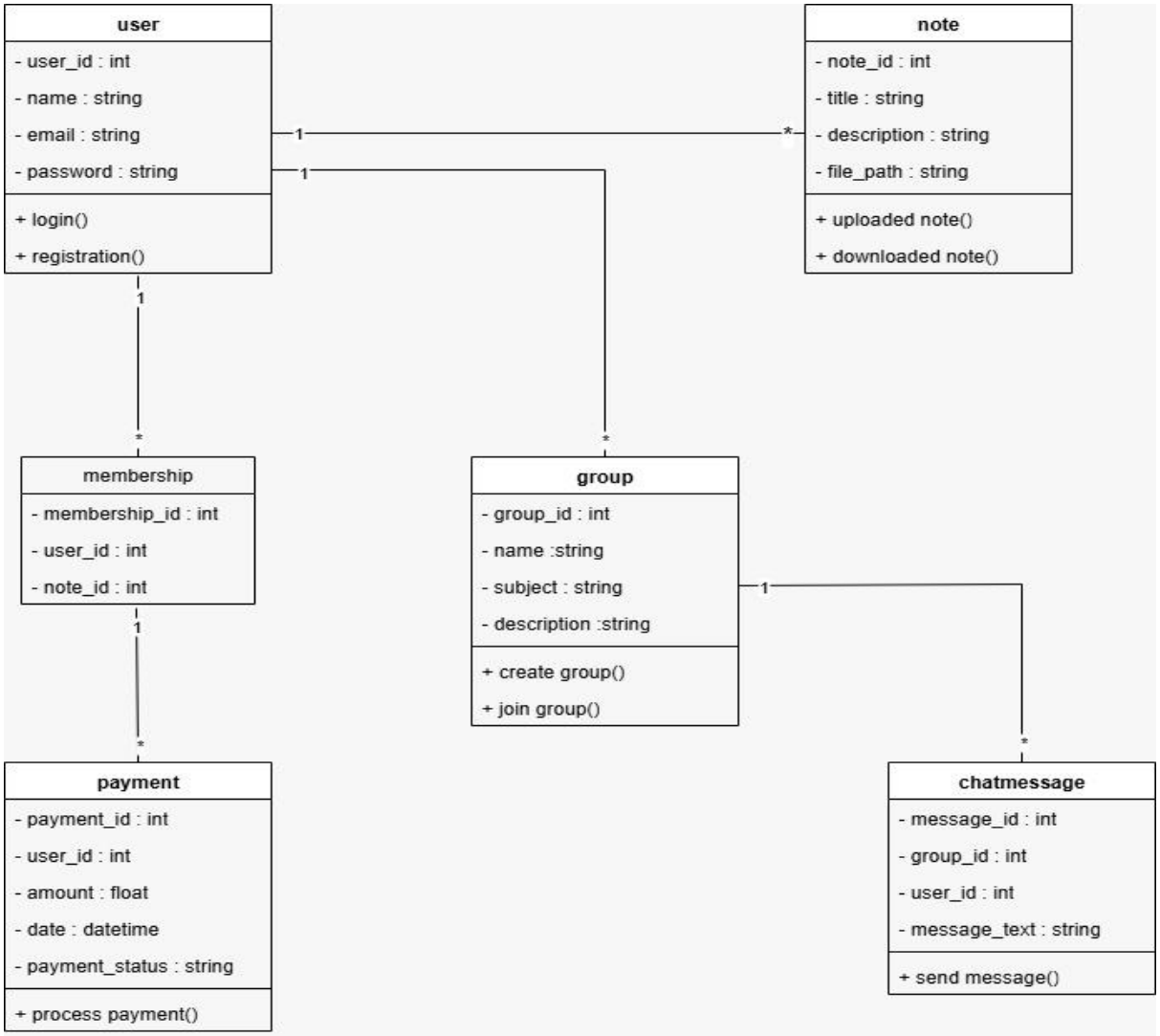


Payment


WEB/APP UI
Auth Service
User DB
Paymen Getway
Start Payment
verify user session
Check user record
return verification
send verified result
Send payment request(amount , user id, item id)
Return payment status(Success/failure)
Store transaction details(user id,amount,status)
Confirm data saved
Display payment result

Class diagram

A Class Diagram is a type of UML (Unified Modelling Language) diagram that shows the static structure of a system by representing its classes, attributes, methods, and relationships. It is mainly used in object-oriented design to describe how different classes interact with each other.



Data dictionary

A Data Dictionary is a detailed description of all the data elements used in a database or information system. It defines each data item's name, type, size, format, and purpose, along with its relationships with other data elements.

User:

Table name	User
Primary key	User_id

Field name	**Data type**	**size**	**description**	**constraints**
User_id	INT	11	Unique ID for each user	Primary key, Auto Increment
name	VARCHAR	30	User's full name	Not Null
Email	VARCHAR	20	User's email address	Unique , Not Null
password	VARCHAR	10	Encrypted password	Not Null

2. Group table

Table name	Group
Primary key	Group_id
Foreign key	User_id

Field name	Data type	size	Description	Constraints
Group_id	INT	11	Unique ID for each group	Primary key , Auto Increment
User_id	INT	11	Id of user	Foreign key
name	VARCHAR	30	Study Group name	Not Null
subject	VARCHAR	20	Subject / topic for group	Not Null
description	TEXT	-	Details about the group	Nullable

3. Membership table

Table name	Membership
Membership_id	Primary key
Group_id	Foreign key
User_id	Foreign key

Field name	Data type	Size	Description	Constraint
Membership_id	INT	11	Unique membership ID	Primary key , Auto increment
Group_id	INT	11	ID of User who joined	Foreign key
User_id	INT	11	ID of group joined	Foreign key

4. ChatMessage table

Table name	ChatMessage
Message_id	Primary key
User_id	Foreign key
Group_id	Foreign key

Field name	Data type	Size	Description	Constraint
Message_id	INT	11	Unique ID for each message	Primary key , Auto increment
User_id	INT	11	User who sent the message	Foreign key
Group_id	INT	11	Group in which message sent	Foreign key
Message_text	TEXT	-	Chat message text	Not Null

5. Note Table

Table name	note
Note_id	Primary key

Field name	Data type	Size	Description	Constraints
Note_id	INT	11	Unique ID for each note	Primary key , Auto Increment
title	VARCHAR	30	Title of the note	Not Null
description	TEXT	-	Note description	Nullable
File_path	VARCHAR	20	File storage path	Not Null

6. Payment table

Table name	Payment
Payment_id	Primary key
User_id	Foreign key

Field name	Data type	Size	Description	Constraints
Payment_id	INT	11	Unique payment ID	Primary key , Auto Increment
User_id	INT	11	User making the payment	Foreign key
amount	DECIMAL	10.5	Payment amount	Not Null
date	DATETIME	-	DATE of payment	Default CURRENT_TIMESTAMP
Payment_status	VARCHAR	50	status	Default =“pending”

6. Bibliography :

www.google.com

www.youtube.com